

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
plt.style.use("ggplot")
```

```
df = pd.read_csv("/content/HHS_Unaccompanied_Alien_Children_Program.csv")
```

```
df.head()
```

	Date	Children apprehended and placed in CBP custody*	Children in CBP custody	Children transferred out of CBP custody	Children in HHS Care	Children discharged from HHS Care
0	December 21, 2025	6.0	18.0	11.0	2,484	14.0
1	December 18, 2025	11.0	50.0	6.0	2,472	16.0
2	December 17, 2025	7.0	31.0	11.0	2,481	10.0
3	December 16, 2025	8.0	54.0	15.0	2,468	9.0
4	December 15, 2025	11.0	42.0	9.0	2,470	7.0



Next steps:

[Generate code with df](#)
[New interactive sheet](#)

```
df.columns = df.columns.str.strip()
```

```
df['Date'] = pd.to_datetime(df['Date'])
```

```

for col in df.columns[1:]:
    df[col] = (
        df[col]
        .astype(str)
        .str.replace(',', '')
    )

    df[col] = pd.to_numeric(
        df[col],
        errors='coerce'
    )

df.dropna(inplace=True)

df.info()

```

```

<class 'pandas.core.frame.DataFrame'>
Index: 720 entries, 0 to 719
Data columns (total 6 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Date                                     720 non-null    datetime64[ns]
1   Children apprehended and placed in CBP custody* 720 non-null    float64
2   Children in CBP custody                  720 non-null    float64
3   Children transferred out of CBP custody  720 non-null    float64
4   Children in HHS Care                    720 non-null    float64
5   Children discharged from HHS Care        720 non-null    float64
dtypes: datetime64[ns](1), float64(5)
memory usage: 39.4 KB

```

```
print(df.shape)
```

```
(720, 6)
```

```
df.isnull().sum()
```

	0
Date	0
Children apprehended and placed in CBP custody*	0
Children in CBP custody	0
Children transferred out of CBP custody	0
Children in HHS Care	0
Children discharged from HHS Care	0

dtype: int64

df.describe()

	Date	Children apprehended and placed in CBP custody*	Children in CBP custody	Children transferred out of CBP custody	Children in HHS Care	Children discharged from HHS Care
count	720	720.000000	720.000000	720.000000	720.000000	720.000000
mean	2024-07-06 05:29:59.999999744	93.523611	171.494444	128.668056	6061.275000	173.406944
min	2023-01-12 00:00:00	0.000000	7.000000	0.000000	1972.000000	0.000000
25%	2023-10-16 18:00:00	12.000000	36.000000	14.000000	2467.750000	19.750000
50%	2024-07-05 12:00:00	99.000000	193.000000	157.000000	6406.500000	181.000000
75%	2025-03-25 06:00:00	147.250000	263.250000	199.250000	8010.250000	267.000000
max	2025-12-21 00:00:00	333.000000	531.000000	440.000000	11516.000000	505.000000
std	NaN	72.646625	126.354965	97.322012	2833.070109	125.702841

```
df['Year'] = df['Date'].dt.year

df['Month'] = df['Date'].dt.month

df['Weekday'] = df['Date'].dt.day_name()
```

```
df['Transfer Efficiency Ratio'] = (
    df['Children transferred out of CBP custody']
    /
    df['Children apprehended and placed in CBP custody*']
) * 100
```

```
df['Discharge Effectiveness'] = (
    df['Children discharged from HHS Care']
    /
    df['Children in HHS Care']
) * 100
```

```
df['CBP Backlog'] = (
    df['Children apprehended and placed in CBP custody*']
    -
    df['Children transferred out of CBP custody']
)
```

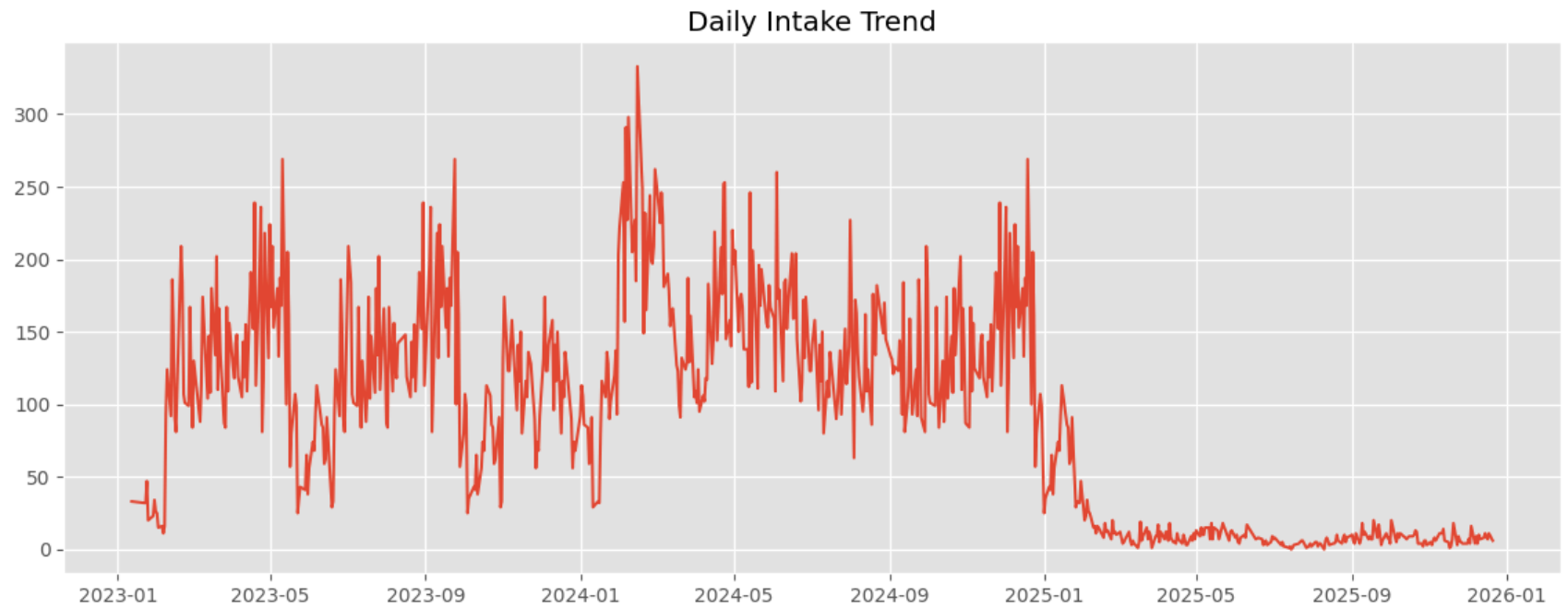
```
df['HHS Backlog'] = (
    df['Children in HHS Care']
    -
    df['Children discharged from HHS Care']
)
```

```
df['Net Flow'] = (
    df['Children apprehended and placed in CBP custody*']
    -
    df['Children discharged from HHS Care']
)
```

```
plt.figure(figsize=(14,5))

plt.plot(
    df['Date'],
    df['Children apprehended and placed in CBP custody*']
)

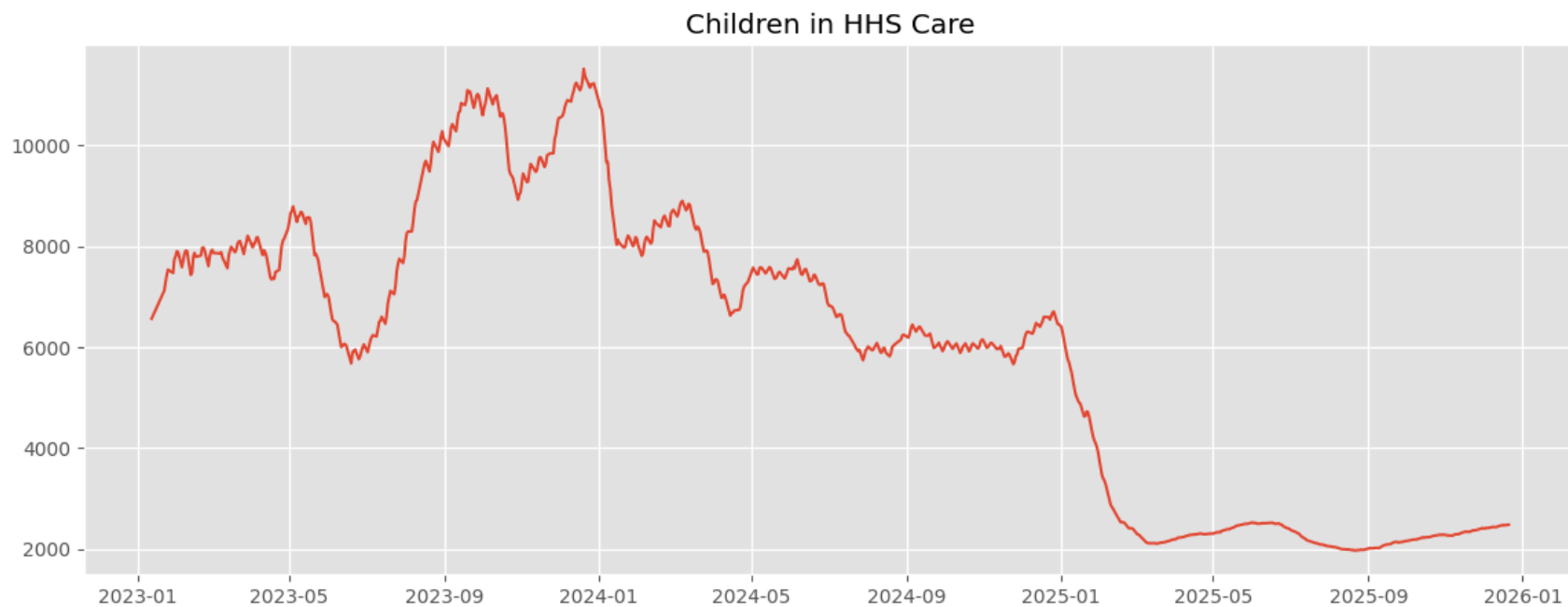
plt.title("Daily Intake Trend")
plt.show()
```



```
plt.figure(figsize=(14,5))

plt.plot(
    df['Date'],
    df['Children in HHS Care']
)

plt.title("Children in HHS Care")
plt.show()
```

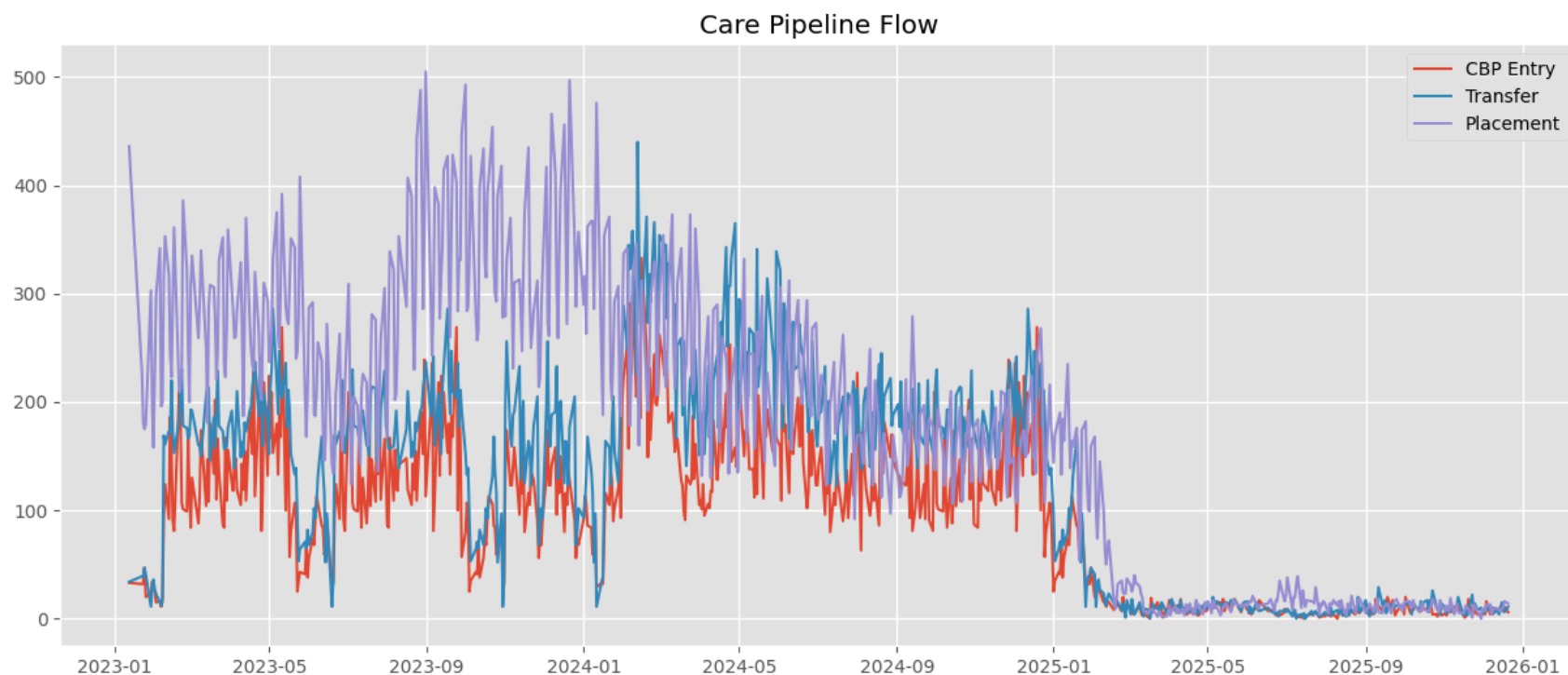


```
plt.figure(figsize=(15,6))

plt.plot(
    df['Date'],
    df['Children apprehended and placed in CBP custody*'],
    label='CBP Entry'
)

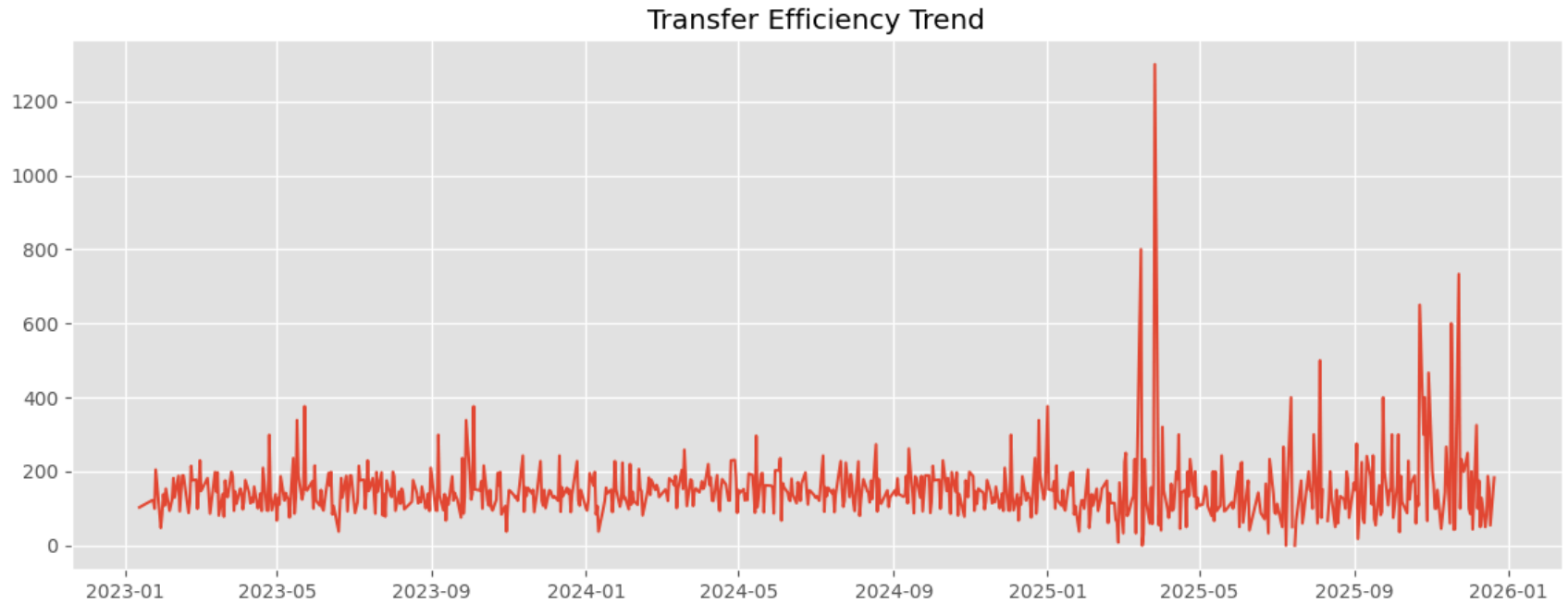
plt.plot(
    df['Date'],
```

```
df['Children transferred out of CBP custody'],  
label='Transfer'  
)  
  
plt.plot(  
    df['Date'],  
    df['Children discharged from HHS Care'],  
    label='Placement'  
)  
  
plt.legend()  
  
plt.title("Care Pipeline Flow")  
  
plt.show()
```

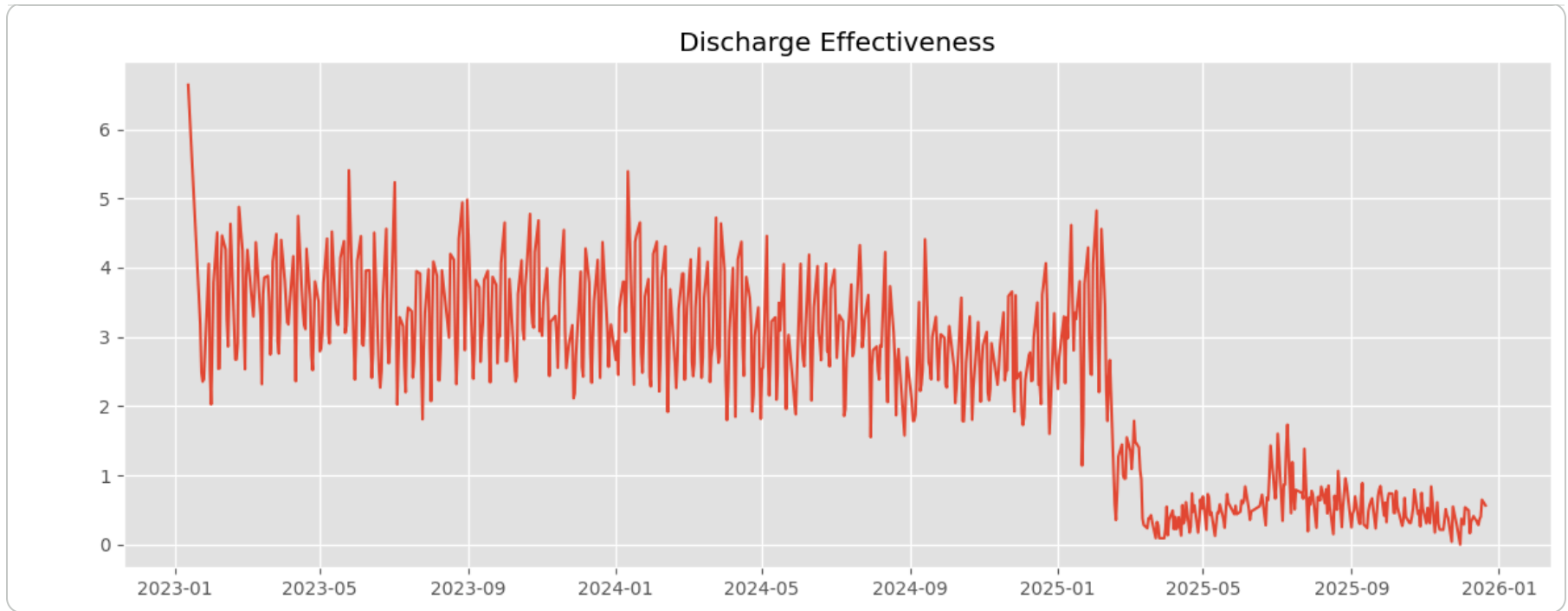


```
plt.figure(figsize=(14,5))
```

```
plt.plot(  
    df['Date'],  
    df['Transfer Efficiency Ratio']  
)  
  
plt.title("Transfer Efficiency Trend")  
  
plt.show()
```



```
plt.figure(figsize=(14,5))  
  
plt.plot(  
    df['Date'],  
    df['Discharge Effectiveness']  
)  
  
plt.title("Discharge Effectiveness")  
  
plt.show()
```

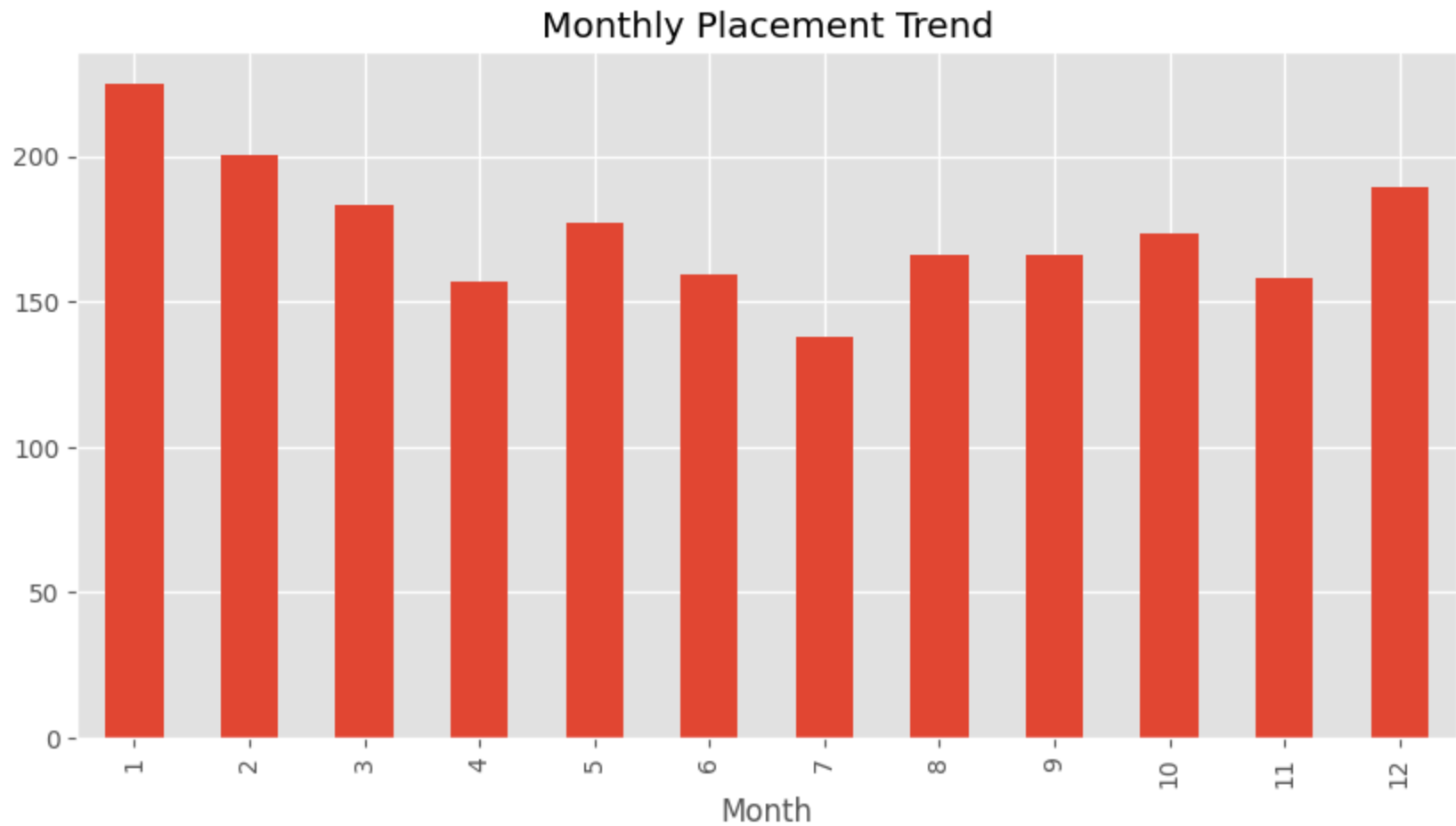



```
monthly = df.groupby('Month')[
    'Children discharged from HHS Care'
].mean()

monthly.plot(
    kind='bar',
    figsize=(10,5)
)

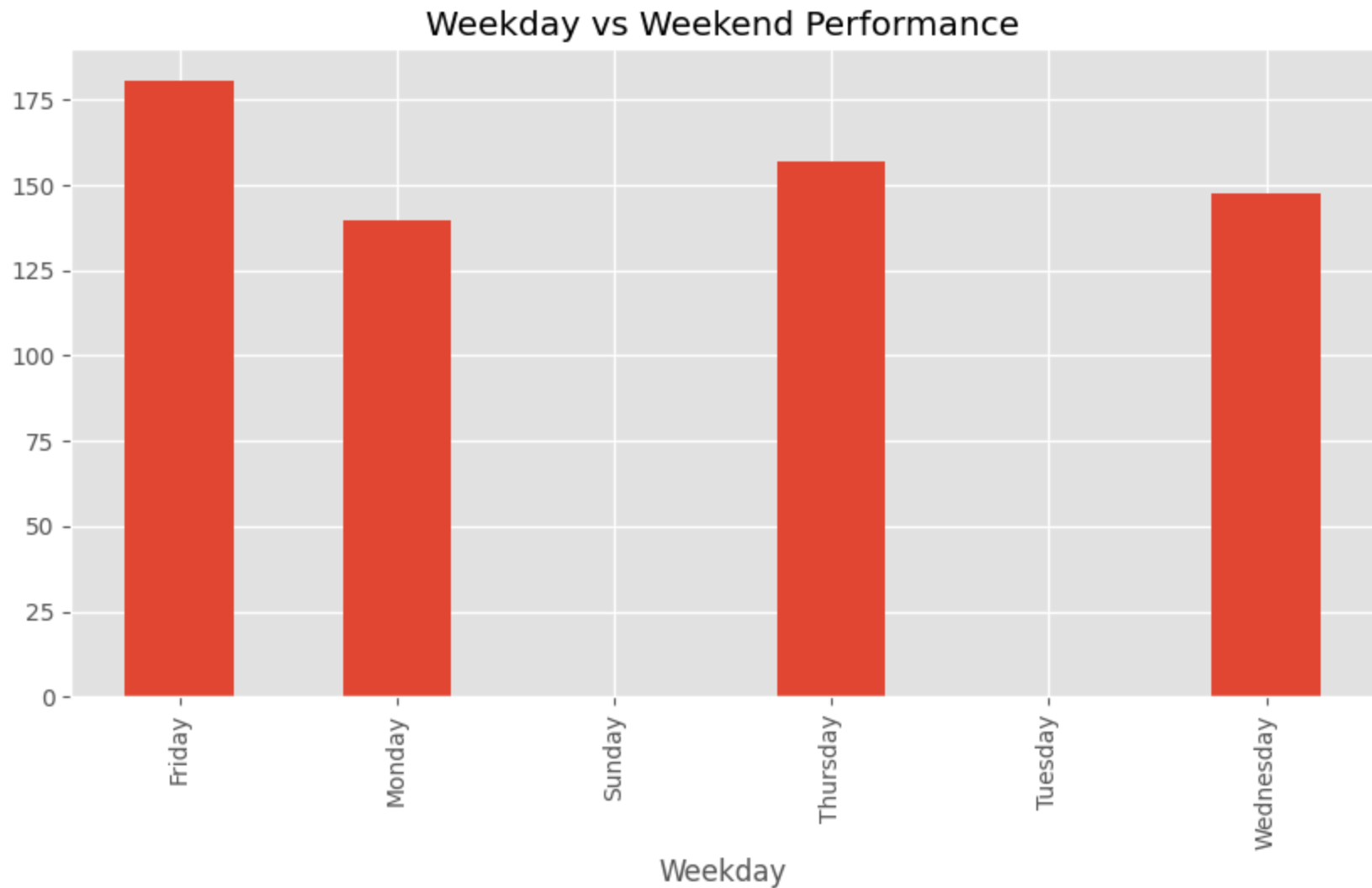
plt.title("Monthly Placement Trend")

plt.show()
```



```
weekday = df.groupby('Weekday')[  
    'Transfer Efficiency Ratio'  
].mean()  
  
weekday.plot(  
    kind='bar',  
    figsize=(10,5)  
)  
  
plt.title("Weekday vs Weekend Performance")
```

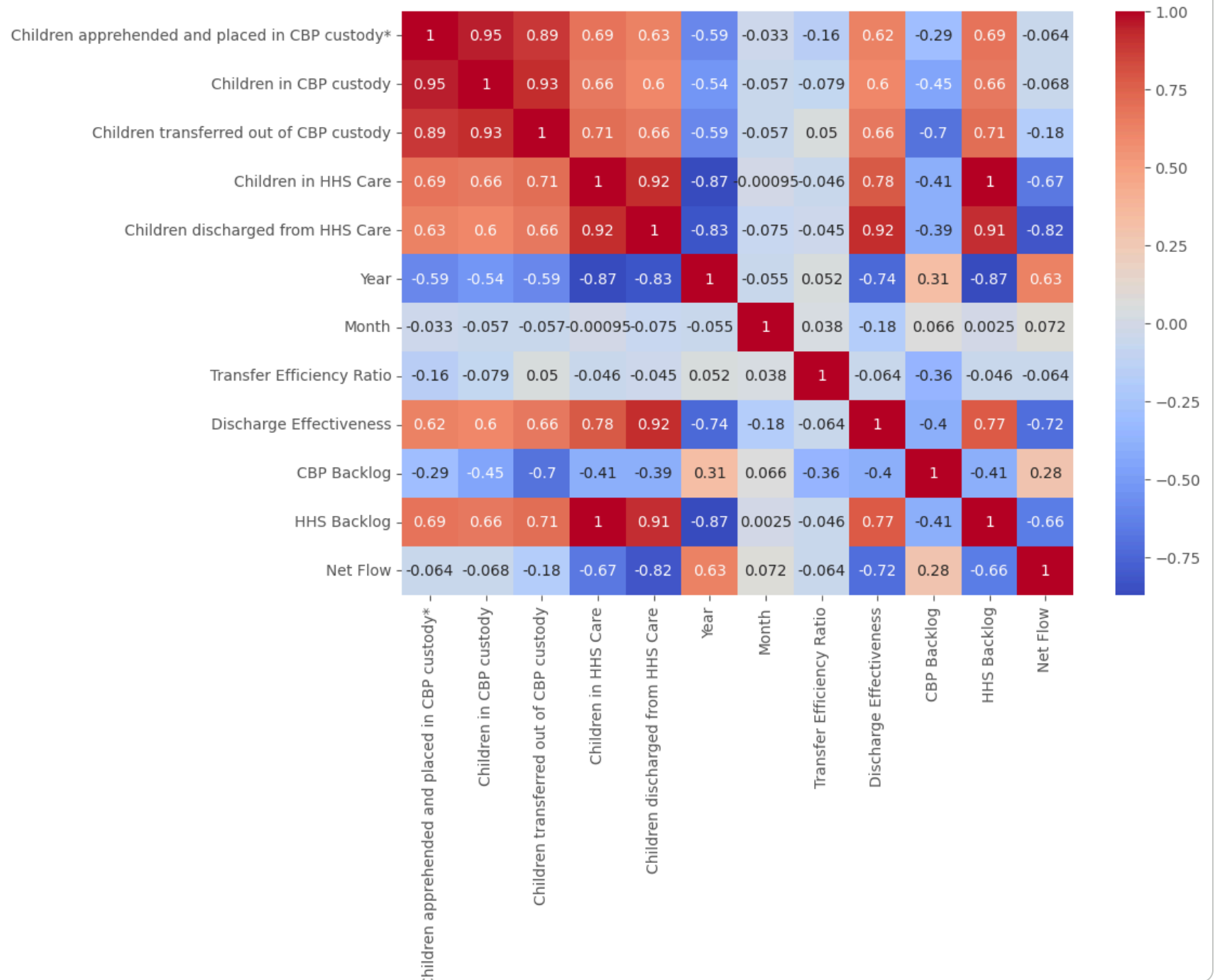
```
plt.show()
```



```
plt.figure(figsize=(10,7))
```

```
sns.heatmap(  
    df.select_dtypes(include='number').corr(),  
    annot=True,
```

```
        cmap='coolwarm'  
    )  
  
plt.show()
```



```
entries = df[
    'Children apprehended and placed in CBP custody*'
].sum()

exits = df[
    'Children discharged from HHS Care'
].sum()

throughput = (exits / entries) * 100

print("Pipeline Throughput Rate:", throughput)
Pipeline Throughput Rate: 185.41515066011257
```

```
df['30 Day Avg'] = (
    df['Children discharged from HHS Care']
    .rolling(30)
    .mean()
)
```